

S-matrix:

$$\langle f | S | i \rangle_{\text{Heisenberg}} = \langle f | j_0 | i \rangle_{\text{Schrodinger}}$$

Cross-sections:

$$\sigma = \pi r^2$$

* Single nucleus,

$$\sigma = \frac{\# \text{ of particles scattered}}{(\text{time})(\text{number density in beam})(\text{velocity of beam})}$$

$$\sigma = \frac{1}{T \Phi} N$$

T: time for experiment

Φ : incoming flux

N: # of particles scattered

$$\Phi = \rho_N |\vec{V}|$$

* Differential Cross Section: $\frac{d\sigma}{d\Omega}$

* # of scattered particles in a certain solid angle $d\Omega$

* Classical diff. prob. $P = \frac{N}{N_{inc}}$

N: Part. scat. into a given area

N_{inc} : incident particles

Q.M. Cross-section: $d\sigma = \left(\frac{1}{T \Phi} \right) dP$

* diff. # of scattering events measured in a collider

$$dN = L \times d\sigma$$

$$L \equiv \frac{dN}{d\sigma}$$

L: Luminosity

$|i\rangle$: two particle state

$2 \rightarrow n$ Process

$$p_1 + p_2 \rightarrow \{p_3\}$$



In the rest frame of one of the particles the magnitude

$$\Phi = \frac{|\vec{v}|}{V}$$

of the flux is just the magnitude of P_2 divided by the total volume

In the C.M. frame beams of particles comes in from both sides

$$\Phi = \frac{|\vec{v}_1 - \vec{v}_2|}{V}$$

$$d\sigma = \frac{V}{T} \frac{1}{|\vec{v}_1 - \vec{v}_2|} dP$$

$$dP = \frac{|\langle f | S | i \rangle|^2}{\langle f | f \rangle \langle i | i \rangle} d\pi$$

$d\pi$: Region of the final state momenta d^3p_j

$$d\pi = \prod_j \frac{V}{(2\pi)^3} d^3p_j \quad (\text{must integrate to } 1)$$

$$\int \frac{dP}{2\pi} = \frac{1}{L} \quad \text{Since} \quad x_i = \frac{iL}{N}$$

$$p_i = \frac{2\pi}{L} \frac{i}{N}$$

$$\langle f | f \rangle \neq 1 \quad \langle i | i \rangle \neq 1 \quad \text{since}$$

$$\langle p | n \rangle = (2\pi)^3 2\omega_p \delta^3(p - n)$$

$$\langle p | p \rangle = (2\pi)^3 2\omega_p \delta^3(0)$$

$$\text{but } (2\pi)^3 \delta^3(0) = \int d^3x e^{i\vec{p} \cdot \vec{x}}$$

$$(2\pi)^3 \delta^3(0) = V$$

$$\Rightarrow \langle p | p \rangle = 2E_p V$$

$$|i\rangle = |p_1\rangle \otimes |p_2\rangle$$

$$\langle f | = \langle p_j |$$

$$\langle i | i \rangle = \langle p_2 | p_2 \rangle \langle p_1 | p_1 \rangle = 4E_{p_1} E_{p_2} V^2$$

$$\langle f|f \rangle = \prod_j (2E_j V)$$

$$S = \mathbb{1} + i\mathcal{T}$$

L
free

$\mathcal{T}$: transfer matrix

$$S = e^{i\mathcal{T}}$$

$$\mathcal{T} = (2\pi)^4 \delta^4(\sum_m p) \mathcal{M}$$

$$\delta^4(\sum_i p_i^\mu - \sum_f p_f^\mu)$$

Matrix elements: $\langle f|M|i \rangle$

$$\langle f|S-i\mathbb{1}|i \rangle = i(2\pi)^4 \delta^4(\sum p) \langle f|M|i \rangle$$

$$|\langle f|S|i \rangle|^2 = |\langle f|\mathbb{1} + i\mathcal{T}|i \rangle|^2$$

$$= \langle f|\mathbb{1} + i\mathcal{T}|i \rangle (\langle f|\mathbb{1} - i\mathcal{T}|i \rangle)^*$$

$$= |\langle f|i \rangle|^2 + |\langle f|\mathcal{T}|i \rangle|^2$$

$$= |\langle f|i \rangle|^2 + (2\pi)^8 (\delta^4(\sum p))^2 |\langle f|M|i \rangle|^2$$

• One delta δ^4 will always yield an infinite volume and so momentum conservation $\delta^4(0)$

$$\delta^4(0) = \frac{TV}{(2\pi)^4}$$

$$|\langle f|i \rangle|^2 = 0?$$

$$|M|^2$$

$$|||$$

$$|\langle f|S|i \rangle|^2 = (2\pi)^4 TV \delta^4(\sum p) |M|^2$$

$$dP = \frac{|\langle f|S|i \rangle|^2}{\langle f|f \rangle \langle i|i \rangle} \prod_j \frac{\pi V}{(2\pi)^3} d^3 p_j$$

$$\hookrightarrow dP = \left(\frac{(2\pi)^4 TV \delta^4(\sum p) |M|^2}{(2E_1 V)(2E_2 V) \prod_k (2E_k V)} \right) \prod_j \frac{\pi V}{(2\pi)^3} d^3 p_j$$

$$dP = (T \prod_j 1 \prod_k 1 \prod_l 1 \prod_m 1 \prod_n 1 \prod_o 1 \prod_p 1 \prod_q 1 \prod_r 1 \prod_s 1 \prod_t 1 \prod_u 1 \prod_v 1 \prod_w 1 \prod_x 1 \prod_y 1 \prod_z 1 \prod_{\dots} 1)$$

$$d\Gamma = \left(\frac{1}{V}\right) \frac{1}{2E_1 2E_2} |M|^2 d\mathcal{V}_{LIPS}$$

$$d\mathcal{V}_{LIPS} = \prod_{\text{final states } j} \frac{d^3 p_j}{(2\pi)^3} \frac{1}{2E_j} (2\pi)^4 \delta^4(\Sigma p)$$

(Lorentz Invariant Phase Space)

$$d\sigma = \frac{V}{T} \frac{1}{|\vec{v}_1 - \vec{v}_2|} dP$$

$$d\sigma = \frac{|M|^2}{2E_1 2E_2 |\vec{v}_1 - \vec{v}_2|} d\mathcal{V}_{LIPS}$$

Decay Rates: prob. that a one-part. state w mom p_1 turns into a multiparticle state w momenta $\{p_i\}$ over a time T

$$d\Gamma = \frac{1}{T} dP$$

$1 \rightarrow n$

$$d\Gamma = \frac{1}{2E_1} |M|^2 d\mathcal{V}_{LIPS} \quad (\text{Rest frame})$$

ex: $2 \rightarrow 2$ Scattering in the C.O.M frame

$$\left\{ \begin{array}{l} \vec{p}_1 = -\vec{p}_2 \\ \vec{p}_3 = -\vec{p}_4 \end{array} \right\} \quad p_1 + p_2 \rightarrow p_3 + p_4$$

$$E_1 + E_2 = E_3 + E_4 = E_{CM}$$

$$d\mathcal{V}_{LIPS} = (2\pi)^4 \delta^4(\Sigma p) \frac{d^3 p_3}{(2\pi)^3} \frac{1}{2E_3} \frac{d^3 p_4}{(2\pi)^3} \frac{1}{2E_4}$$

• Integrate over p_4 using δ

$$\delta^4(p_3 - p_4) = \int d^4 p_4 e^{i(p_3 - p_4) \cdot x} \dots$$

$$\frac{1}{(2\pi)^4} \int d^4 p_4 e^{i(p_3 - p_4) \cdot x} = \delta^4(p_3 - p_4)$$

$$\therefore \int d\pi_{LIPS} = 16\pi^2 \frac{d\Omega}{4\pi} \int dP_f \frac{1}{E_3 E_4} \delta(E_3 + E_4 - E_{cm})$$

$$P_f = |P_3| = |P_4|, \quad E_i = \sqrt{m_i^2 + P_f^2}$$

$$P_f \rightarrow X(P_f) \equiv E_3(P_f) + E_4(P_f) - E_{cm}$$

Jacobian: $dx = \frac{dx}{dP_f} dP_f = \frac{E_3 + E_4}{E_3 E_4} P_f dP_f$

$$\int d\pi_{LIPS} = \frac{1}{16\pi^2} (d\Omega) \int_{m_3+m_4-E_{cm}}^{\infty} dx P_f \frac{E_3 E_4}{E_{cm} E_3 E_4} \delta(x)$$

$$\int d\pi_{LIPS} = \left(\frac{d\Omega}{16\pi^2} \right) \int_{m_3+m_4-E_{cm}}^{\infty} dx \left(P_f / E_{cm} \right) \delta(x)$$

$$= \frac{1}{16\pi^2} d\Omega \frac{P_f}{E_{cm}} \Theta(E_{cm} - m_3 - m_4)$$

$$d\sigma = \frac{1}{2E_1 2E_2 |v_1 - v_2| 16\pi^2} d\Omega \frac{P_f}{E_{cm}} |\mathcal{M}|^2 \Theta(E_{cm} - m_3 - m_4)$$

$$|v_1 - v_2| = \left| \frac{|P_1|}{E_1} + \frac{|P_2|}{E_2} \right| = |P_1| \frac{E_{cm}}{E_1 E_2}$$

$$\left(\frac{d\sigma}{d\Omega} \right)_{cm} = \left(\frac{1}{64\pi^2 E_{cm}^2} \right) \frac{|P_f|}{|P_1|} |\mathcal{M}|^2 \Theta(E_{cm} - m_3 - m_4)$$

$2 \rightarrow 2$ Scattering

* For equal masses $\Theta = 1$, $|P_f| = P_1$

$$\left(\frac{d\sigma}{d\Omega} \right)_{cm} = \frac{1}{64\pi^2 E_{cm}^2} |\mathcal{M}|^2$$